

INTELLIDEN - UX Architecture & Design Deliverables

This document outlines the Sitemaps, User Flows, Storyboards, and Sketch requirements for the **INTELLIDEN** smart home ecosystem across three primary platforms: Mobile/Tablet, AR Headset, and Smartwatch. The architecture is built around the primary persona defined in the project brief.

Primary Persona Context

Marcus (Specialized Automation Engineer)

Marcus is highly technical, deeply passionate about home automation, and lives with roommates or family. He requires high-level data (power draw, sensor readings), advanced automation logic, spatial diagnostics, and seamless conflict resolution mechanisms when other users override his smart home settings.

1. Mobile Phone / Tablet Interface

The mobile and tablet applications serve as the primary "Command Center" for the INTELLIDEN ecosystem. They provide deep configuration, long-term statistics, and complex rule-setting capabilities.

1.1. Sitemap

- **Dashboard:** Quick Toggles, Active Conflicts/Alerts, Energy Consumption Graph.
- **Devices:** Sorted by Room (Living Room, Kitchen, etc.), Add New Device, Device Settings.
- **Automations:** Rule Builder (If This Then That), Gesture Configurations, Schedule Manager.
- **Co-op & Family:** User Permissions, Conflict Resolution Priorities, Guest Access.

1.2. User Flow: Conflict Resolution (Co-op)

Step	User Action	System Response
1	Marcus receives a push notification on his phone.	System detects contradictory input (e.g., Roommate sets thermostat to 25°C, overriding Marcus's 21°C rule).
2	Marcus opens the "Active Conflicts" tab on the Dashboard.	Displays the conflict: "Ioana modified Living Room Temp".
3	Marcus selects a resolution strategy (Override, Accept, or Compromise).	System applies the chosen logic and updates the smart thermostat immediately.

1.3. Storyboard (Mobile)

1. **Scene 1:** Marcus is at work. His roommate feels cold and manually cranks up the smart thermostat.
2. **Scene 2:** Marcus receives an "Efficiency Alert" on his phone indicating unexpected power draw.
3. **Scene 3:** He opens the INTELLIDEN app, navigates to the Co-op tab, and sees the conflict. He uses the "Compromise" feature to set it to 23°C, saving energy while keeping the roommate warm.

1.4. Sketches / Wireframing Guide

What to draw in Figma: A dark-mode dashboard (Deep Space #0A0A0A) with JetBrains Mono font for temperature readouts. Create a card with a 12px border radius showing the "Conflict Alert" in Warning Amber (#FFB800). Include a bottom navigation bar with icons for Dashboard, Rooms, Automations, and Profile.

2. AR Headset (Mixed Reality) Interface

The AR interface is designed for spatial computing, focusing on real-time diagnostics, X-ray views of appliances, and gesture-based interactions without obstructing the physical world.

2.1. Sitemap (Spatial UI)

- **Persistent HUD:** Minimalist status indicators (Current Room Temp, Wi-Fi status) in peripheral vision.
- **Spatial Anchors:** Floating panels tied to physical devices (e.g., a panel hovering over the AC unit).
- **Diagnostic Mode:** Deep-dive "X-ray" view for engineering troubleshooting.
- **Virtual Hand Menu:** Opened via a pinch gesture, offering quick scenes (Movie Mode, Focus Mode).

2.2. User Flow: Real-Time Diagnostic of Malfunctioning Appliance

Step	User Action	System Response
1	Marcus puts on AR glasses and looks at a noisy dishwasher.	AR System recognizes the device via spatial mapping and pulls up its digital twin profile.
2	Marcus performs a "Bloom" hand gesture to open the device menu.	System projects a glassmorphism floating UI with current sensor data next to the appliance.
3	Marcus selects "Diagnostic Mode" by pointing.	System highlights the exact failing internal component in Electric Cyan (#00F2FF) using an AR overlay.

2.3. Storyboard (AR)

1. **Scene 1:** Marcus hears a weird noise from his HVAC unit.
2. **Scene 2:** He equips his AR glasses. Looking at the HVAC, a floating panel appears.
3. **Scene 3:** He activates Diagnostic Mode. An X-ray overlay shows a blocked filter. He orders a replacement directly by tapping the holographic prompt.

2.4. Sketches / Wireframing Guide

What to draw in Figma: Simulate a transparent background (or an image of a living room). Design floating UI panels with 20% background blur (Glassmorphism). Icons should have a 2pt stroke and a subtle outer glow to prevent bleeding into real-world textures. Avoid solid backgrounds.

3. Smartwatch Interface

The Smartwatch app is optimized for micro-interactions (under 10 seconds). It relies heavily on complications, quick toggles, and zero-touch gestures.

3.1. Sitemap

- **Watch Face Complications:** Main screen shortcuts (Current Temp, Lock Status).
- **Quick Actions (Scrollable List):** Unlock Door, Lights Off, Run "Goodnight" Routine.
- **Actionable Notifications:** Alerts that require a simple Yes/No or gesture to dismiss.

3.2. User Flow: Zero-Touch Gesture Unlock

Step	User Action	System Response
1	Marcus arrives at his front door carrying heavy groceries.	Proximity sensors detect his smartwatch nearby and prepare the lock mechanism.

Step	User Action	System Response
2	Marcus performs a pre-defined double wrist flick gesture.	Watch gyroscope detects the pattern and transmits the unlock signal via Bluetooth/UWB.
3	Marcus enters the home.	Smart door unlocks and watch provides a short haptic feedback (buzz) to confirm.

3.3. Storyboard (Smartwatch)

1. **Scene 1:** Marcus arrives at his apartment door with hands full of bags.
2. **Scene 2:** He cannot reach his keys or phone. He uses a specific wrist gesture.
3. **Scene 3:** The door clicks open automatically. A green checkmark flashes on the watch screen with a haptic vibration.

3.4. Sketches / Wireframing Guide

What to draw in Figma: Extremely minimal, circular or small rectangular screens. High contrast dark mode (pure black background to blend with OLED screen). Large, single buttons per screen.